

Red Torque – Detailed Major Component Study

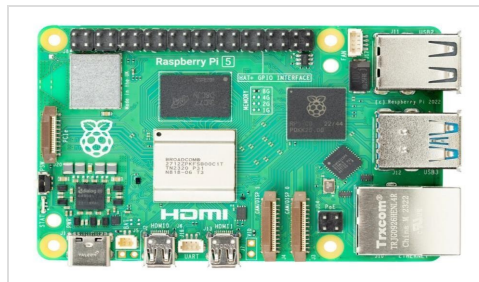
Date: 2 May 2026

1. Raspberry Pi 5

The Raspberry Pi 5 is the latest and most powerful version of the Raspberry Pi computer series. It looks like a small board but has the power of a full desktop. It is made for DIY and educational projects, especially robotics.

In our robot project, it is the main brain handling image processing, decision-making, and the PID wall-following logic, using data forwarded from the sensor system.

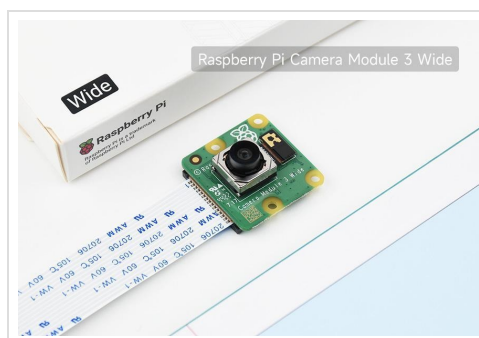
- Quad-core 64-bit ARM Cortex-A76 CPU running at 2.4GHz – faster than Pi 4.
- Dual camera ports – supports two camera modules at the same time.
- PCIe 2.0 interface – allows fast SSD storage or expansion cards.
- Real-time clock – keeps track of time even without power.
- Improved GPIO performance – faster and better I/O for sensors.
- Supports active cooling via fan connector directly on board.
- USB 3.0 ports – faster data transfer from external devices.
- Used as the main controller to run Linux (Raspberry Pi OS).



The Raspberry Pi 5 acts as the main brain of the robot, handling tasks like image processing, sensor integration, PID wall-following, and decision-making. It runs the software that controls navigation, processes camera input, and communicates with the two ESP32 boards.

2. Raspberry Pi Camera Module 3 Wide

The Raspberry Pi Camera Module 3 Wide is an advanced camera designed specifically for Raspberry Pi boards, offering improved image quality and a wider field of view for embedded vision applications.



- Provides up to 12MP resolution using the Sony IMX708 sensor, supporting high-quality image capture and video recording.
- Offers a 120° wide field of view, making it ideal for applications requiring broader scene coverage such as robotics navigation and surveillance.
- Supports HDR (High Dynamic Range) for improved performance in varying lighting conditions.
- Includes autofocus capability, enabling sharp image capture without manual lens adjustment.
- Connects via the CSI (Camera Serial Interface) port, ensuring high-speed, low-latency data transfer compared to USB webcams.

- Fully compatible with OpenCV, **picamera2/libcamera**, and Python-based image processing workflows.
- Suitable for real-time applications such as object detection, lane/line following, and spatial awareness in robotic systems.

In our robot, this camera connects directly to the Raspberry Pi 5's CSI port and is used to detect red and green cubes, providing input for navigation decisions. It replaced an earlier USB webcam to take advantage of the faster, lower-latency CSI interface.

3. ESP32 Microcontroller

The ESP32 is a low-cost, low-power microcontroller board that includes built-in Wi-Fi and Bluetooth. It is popular in IoT and robotics projects.

- Dual-core processor with speeds up to 240 MHz.
- Wi-Fi and Bluetooth capabilities make it suitable for wireless communication.
- Can be programmed using Arduino IDE or MicroPython.
- Multiple GPIO pins to connect sensors and control motors.
- I2C, SPI, UART interfaces for communication with other boards and modules.



Our project uses **two identical ESP32 boards**, each given a different job so neither one becomes a bottleneck:

- **ESP32 #1 – Communication Bridge:** sits between the Raspberry Pi 5 (USB serial) and the LEGO SPIKE Prime Hub (UART). It relays motor and steering commands from the Pi to the Hub, and passes Hub status back to the Pi. It does not make any navigation decisions – and never did; this was clarified early in the project after some initial documentation briefly suggested otherwise.
- **ESP32 #2 – Sensor Relay:** dedicated purely to sensors. Reads the two TF-Luna units – each on its own separate I2C bus – and the front TF-LC02 over UART1, then relays all three readings to the Raspberry Pi 5 over USB. This board was added specifically to solve a TF-Luna address-conflict problem (see Section 4).

Splitting sensor reading and bridging duties across two boards avoids overloading a single microcontroller with both real-time sensor polling and time-critical motor/steering command relay.

4. TF-Luna LiDAR Sensor (×2 – Left & Right)

The TF-Luna is a compact, low-cost Time-of-Flight (ToF) LiDAR sensor used for short-range distance measurement. Two units are mounted on the left and right sides of the robot chassis.



TF-Luna ToF LiDAR Module

- Single-point ranging LiDAR based on the ToF (Time-of-Flight) principle.

- Detection range: 0.2m – 8m, with 1cm distance resolution.
- Supports both UART and I2C communication interfaces – our build uses I2C mode, enabled by grounding the sensor's mode-select pin.
- Compact and lightweight, easy to mount on a LEGO Technic chassis.
- Operates on a 5V supply with 3.3V communication logic levels.
- Used in our project for continuous, real-time wall-distance measurement, feeding a PID control loop running on the Raspberry Pi to keep the robot centered between walls – the sensor's narrow beam suits this continuous side-facing measurement well.

Both TF-Luna units connect to ESP32 #2, which relays their readings to the Raspberry Pi 5 for PID processing. Since both units share the same factory-default I2C address, and changing one unit's address in firmware never reliably stuck, each TF-Luna is instead wired to its own separate I2C bus on ESP32 #2 – sidestepping the address conflict entirely, since neither sensor ever needs a non-default address. These sensors replaced the project's original ultrasonic (HC-SR04) sensors, giving faster, more precise, and more consistent distance readings for wall-following.

5. TF-LC02 LiDAR Sensor (×2 – Front & Rear)

The TF-LC02 is a short-range ToF LiDAR sensor, well suited for close-proximity, high-precision distance detection. Two units are mounted at the front and rear of the robot.



TF-LC02 ToF LiDAR Module

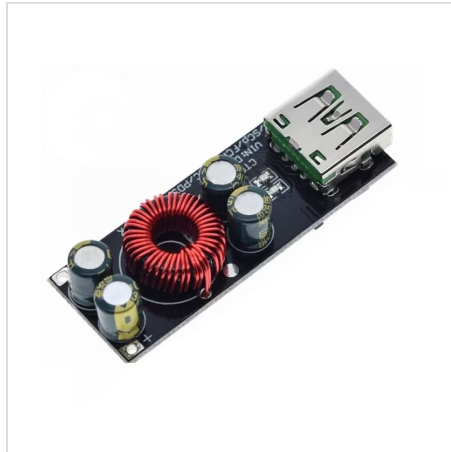
- Single-point ranging ToF LiDAR using a VCSEL (Vertical-Cavity Surface-Emitting Laser) light source.
- Detection range: 3cm – 200cm, ideal for close-range and precision tasks.
- Operates on 3.3V only – must never be connected to a 5V rail.
- Compact form factor, suitable for tight mounting positions on the chassis.
- Compatible with Arduino, Raspberry Pi, and STM32-based platforms.

The **front** TF-LC02 connects to ESP32 #2 over UART1, alongside the two TF-Luna units, and is used for close-range obstacle detection. Because it's fixed in place rather than servo-mounted, it can misread distances mid-turn as the robot rotates into a corner; rather than add a servo mount, this is handled in software – the robot briefly trusts the front sensor less while actively turning. The **rear** TF-LC02 connects directly to the Raspberry Pi 5 through a TTL-USB adapter (no ESP32 involved) and is used for the parking-precision round specifically. Together with the TF-Luna pair, these four LiDAR sensors fully replace the earlier six-sensor ultrasonic (HC-SR04) array used in the previous design.

6. Buck Converter (USB QC4.0/PD3.0 Module)

A buck converter is a voltage step-down converter that changes a high input voltage to a lower output voltage. It is crucial for safely powering electronics from batteries.

- Converts the battery's higher voltage down to the levels needed by onboard electronics.
- Comes with a USB output for easy powering of USB-fed devices.
- Ensures consistent and safe voltage levels to prevent board damage.



3518 Chip Type-C QC4.0/PD3.0 USB Buck Module

The buck converter used in our build is a **3518-chip Type-C USB QC4.0/PD3.0 Quick Fast Charge Adapter Module**, with a wide 6V–35V input range. Its onboard toroidal inductor and output USB-A port trigger fast-charge voltages using the Quick Charge 4.0 and Power Delivery 3.0 protocols, making it a convenient way to draw a clean, regulated voltage straight off the battery pack.

- Input range: 6V–35V, comfortably covering the robot's 7.4V LiPo battery.
- Supports QC4.0 and PD3.0 fast-charge negotiation over a USB-A output port.
- Built around the 3518 controller chip, with a toroidal inductor for efficient, stable conversion.
- Used in our build to power USB-fed components (including the Raspberry Pi, ESP32 boards, and other USB-powered electronics) directly and efficiently from the battery pack.

The TF-LC02 sensors, being 3.3V-only, are powered from a separately regulated 3.3V rail, kept isolated from the 5V line to avoid damage.

7. Battery

The GenX Power Premium 7.4V LiPo Battery is a high-performance rechargeable battery commonly used in robotics, drones, and embedded systems requiring stable and high current output.



- Available in multiple capacities such as 650mAh, 1000mAh, 2200mAh, up to 8000mAh, allowing flexibility based on power requirements.
- Operates at a nominal voltage of 7.4V (2S configuration), meaning two cells connected in series for stable output.
- Provides high discharge rates (25C–80C), enabling it to supply large currents required for motors, drivers, and high-load components.
- Lightweight and compact design with high energy density, making it ideal for mobile robots and UAV systems.
- Typically comes with connectors like XT60, ensuring reliable and low-resistance power delivery.
- Built using matched cells for consistent performance and longer usable runtime per charge cycle.
- Suitable for applications such as robotics, RC vehicles, drones, and embedded projects requiring portable power.

8. LEGO SPIKE Prime Set (Chassis & Command Executor)

The LEGO SPIKE Prime (Chassis & Hub) is an integrated robotics platform that combines a modular mechanical structure with a programmable control hub. In our system, the SPIKE Prime Hub is a pure command executor – it receives drive/steer commands from ESP32 #1 and runs them directly, with no PID or path-planning logic of its own. All navigation decisions are made on the Raspberry Pi; the Hub's job is simply to execute them.



- Consists of over 500+ LEGO Technic elements, enabling construction of a customizable and reconfigurable robot chassis.
- Includes a programmable Intelligent Hub (microcontroller) based on STM32 architecture, responsible for direct motor/steering execution.
- The Hub features 6 input/output ports, allowing connection of motors and sensors for full robotic control.
- Has a **built-in 6-axis IMU (gyro + accelerometer)** for orientation and angular turn tracking – this comes included with the Hub itself, so no separate IMU module is used in our build.
- Also includes a 5×5 LED matrix, Bluetooth connectivity, and a speaker for feedback and interaction.
- Supports both block-based (Scratch) and Python programming; our project runs on **Pybricks**.
- The chassis uses LEGO Technic beams, gears, wheels, and connectors, providing a strong mechanical structure for mobility.
- Comes with motors (1 large + 2 medium), used for drive and steering.
- The kit's **Color Sensor** is mounted on the robot facing straight down at the floor, used to detect the orange and blue lap-marking lines on the play field.
- Designed for modular robot design, allowing quick iteration of wheeled robots and autonomous systems.



LEGO SPIKE Prime Color Sensor

In our robot, the Color Sensor from the kit is mounted underneath the chassis, facing the floor, positioned to pass directly over the field's lap-marking lines. It distinguishes the orange line and the blue line from the plain floor color, and each crossing is counted as a lap marker – letting the robot track completed rounds during the Open and Obstacle Challenge runs. It connects to one of the Hub's remaining I/O ports and is read using Pybricks; since lap counting is a simple color-threshold check, this logic runs on the Hub itself alongside motor and steering execution, rather than adding extra load to the Raspberry Pi's vision and PID pipeline.

Note: The Hub's onboard sensor ports were previously earmarked for the robot's distance sensors, with wall-following logic originally planned to run on the Hub itself. Once the Hub ran out of free ports for the four new LiDAR sensors, those sensors – and the PID logic that depends on them – moved to the Raspberry Pi / ESP32 sensor pipeline instead, leaving the Hub to focus purely on motor, steering, and IMU-based execution.

Summary

The core components explored include the Raspberry Pi 5, which functions as the main processor responsible for image processing, decision-making, and PID wall-following logic. The Raspberry Pi Camera Module 3 Wide is used for capturing images via the CSI interface, making it suitable for vision-based cube detection tasks.

Two ESP32 microcontrollers are used in the system: one acts purely as a communication bridge between the Raspberry Pi and the LEGO SPIKE Prime Hub, while the other acts as a dedicated sensor relay, reading the two TF-Luna units – each on its own I2C bus – and the front TF-LC02 sensor over UART1, forwarding all readings to the Pi over USB. A rear TF-LC02 sensor connects directly to the Raspberry Pi through a TTL-USB adapter for parking precision. Together, these four LiDAR sensors (2× TF-Luna, 2× TF-LC02) replace the project's earlier six-sensor ultrasonic array, providing faster and more accurate wall-following, obstacle detection, and parking precision.

The LEGO SPIKE Prime Hub is a pure command executor, running motor and steering commands sent from the Pi and providing orientation data through its built-in 6-axis IMU – no separate IMU module is used, and no PID logic runs on the Hub itself. The kit's Color Sensor, mounted facing the floor, detects the orange and blue lap lines on the field so the robot can count completed rounds. A 7.4V LiPo battery serves as the main power source, supplying energy to all components, with voltage regulation managed through the QC4.0/PD3.0 buck module that safely steps down higher voltage to the levels required by sensitive electronics, including a dedicated 3.3V rail for the TF-LC02 sensors.

Each component plays a vital role in building a stable and efficient electronic system. Their individual functions and purposes are clearly identified and will be used collectively in the system's design and integration.